

INSTRUMENT TRANSFORMERS

Instrument transformers are used in connection with measurement of voltage, current, energy and power in ac circuits.

Two reasons for use of instrument transformers in measurement:

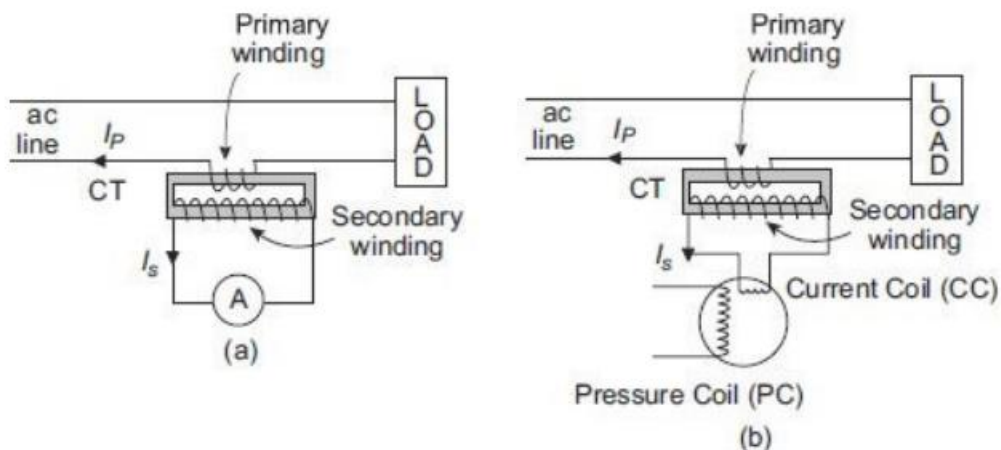
1. to extend (multiply) the range of the measuring instrument
2. to isolate the measuring instrument from a high-voltage line.

Instrument transformers can be effectively used to step down the voltage and current within range of the existing measuring instruments of moderate size.

Instrument transformers are either :-

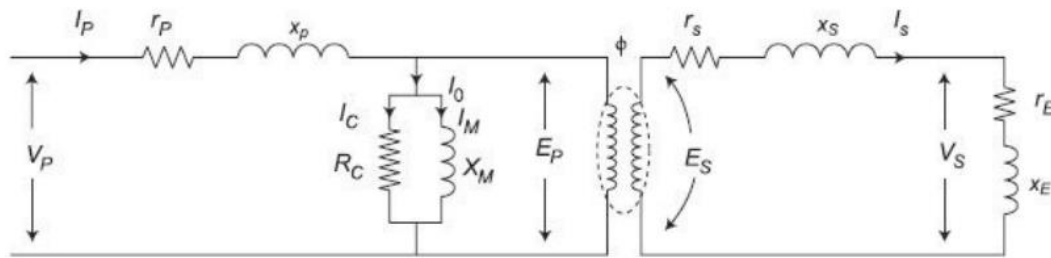
- (a) current transformer or CT,
- (b) voltage or potential transformers or PT.

1. CURRENT TRANSFORMER:

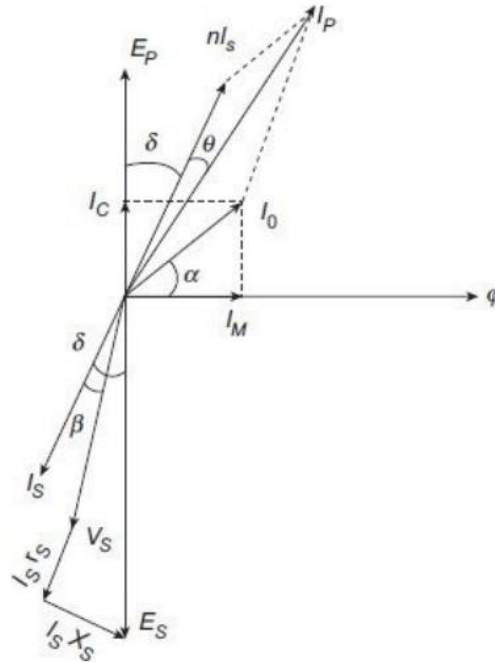


1 CT for (a) current, and (b) power measurement

When a typical name plate rating of a CT shows 500/1 A 5 VA 5P20 it indicates that the CT rated primary and secondary currents are 500 A and 1 A respectively, its rated secondary burden is 5 VA, it is designed to have 5% accuracy and it can carry up to 20 times higher current than its rated value while connected in line to detect fault conditions, etc.



2 Equivalent circuit of a CT



3 Phasor diagram of a CT

θ = the “phase angle” of the CT

δ = phase angle between secondary winding induced voltage and secondary winding current (i.e. phase angle of total burden, including secondary winding)

β = phase angle of secondary load (burden) circuit only

α = phase angle between no-load current I_0 and flux ϕ

V_P = primary supply voltage

E_P = primary winding induced voltage

V_S = secondary terminal voltage

E_S = secondary winding induced voltage

I_P = primary current

I_S = secondary current

I_0 = no-load current

I_C = core loss component of current

I_M = magnetising component of current

r_P = resistance of primary winding

x_P = reactance of primary winding

r_S = resistance of secondary winding

x_S = reactance of secondary winding

R_C = imaginary resistance representing core losses

X_M = magnetising reactance

r_e = resistance of external load (burden) including resistance of meters, current coils,

etc.

x_E = reactance of external load (burden) including reactance of meters, current coils,

etc.

N_P = primary winding number of turns

N_S = secondary winding number of turns

n = turns ratio = N_S/N_P

ϕ = working flux of the CT

θ = the “phase angle” of the CT

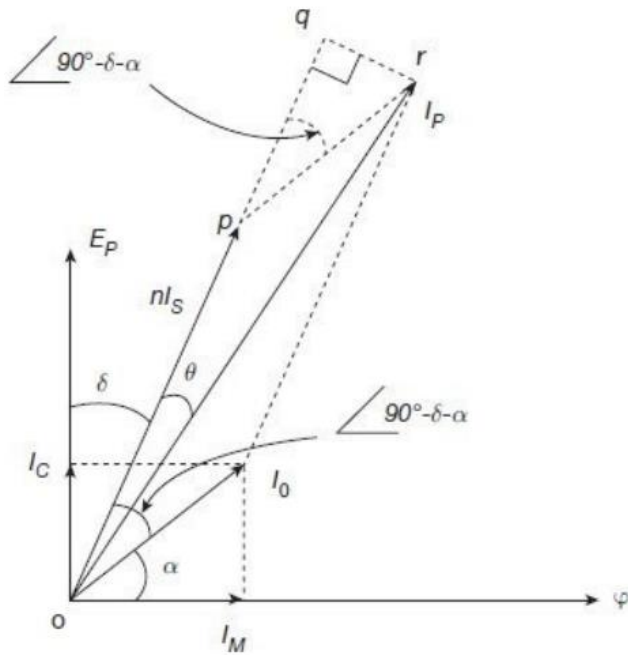
δ = phase angle between secondary winding induced voltage and secondary winding

current (i.e. phase angle of total burden, including secondary winding)

β = phase angle of secondary load (burden) circuit only

α = phase angle between no-load current I_0 and flux ϕ

CURRENT TRANSFORMATION RATIO OF CT:



From the right-angle triangle pqr , we get

$$pr = I_0$$

$$pq = I_0 \cdot \cos(90^\circ - \delta - \alpha) = I_0 \cdot \sin(\delta + \alpha)$$

$$qr = I_0 \cdot \sin(90^\circ - \delta - \alpha) = I_0 \cdot \cos(\delta + \alpha)$$

Now, $(or)^2 = (op + pq)^2 + (qr)^2$

Or, $(I_p)^2 = (nI_s + I_0 \cdot \sin(\delta + \alpha))^2 + (I_0 \cdot \cos(\delta + \alpha))^2$

$$= n^2 I_s^2 + I_0^2 \cdot \sin^2(\delta + \alpha) + 2nI_s I_0 \cdot \sin(\delta + \alpha) + I_0^2 \cdot \cos^2(\delta + \alpha)$$

$$= n^2 I_s^2 + 2nI_s I_0 \cdot \sin(\delta + \alpha) + I_0^2 (\sin^2(\delta + \alpha) + \cos^2(\delta + \alpha))$$

$$= n^2 I_s^2 + 2nI_s I_0 \cdot \sin(\delta + \alpha) + I_0^2$$

$$I_p = \sqrt{n^2 I_s^2 + 2nI_s I_0 \cdot \sin(\delta + \alpha) + I_0^2} \quad (3.1)$$

In a well-designed CT, the no-load current I_0 is much less as compared to the primary current I_p or even the reflected secondary current (which is nominally equal to the primary current) nI_s .

Equation (3.1) can thus now be approximated as

$$I_p = \sqrt{n^2 I_s^2 + 2nI_s I_0 \cdot \sin(\delta + \alpha) + (I_0 \cdot \sin(\delta + \alpha))^2}$$

Hence, $I_p = nI_s + I_0 \cdot \sin(\delta + \alpha)$

CT transformation ratio can now be expressed as

$$R = \frac{I_p}{I_s} = \frac{nI_s + I_0 \cdot \sin(\delta + \alpha)}{I_s} = n + \frac{I_0}{I_s} \sin(\delta + \alpha)$$

Though approximate, Eq. (3.2) is sufficiently accurate for practical estimation of CT transformation ratio. The above equation, however, is true for only when the power factor of the burden is lagging, which is mostly true in all practical inductive meter coils being used as burden.

Equation (3.2) can be further expanded as

$$R = n + \frac{I_0}{I_s} \sin(\delta + \alpha) = n + \frac{I_0}{I_s} (\sin \delta \cos \alpha + \cos \delta \sin \alpha)$$

$$\text{or, transformation ratio } R \approx n + \frac{I_M \sin \delta + I_C \cos \delta}{I_s}$$

[since, $I_M = I_0 \cos \alpha$ and $I_C = I_0 \sin \alpha$]

PHASE ANGLE OF CT:

From the phasor diagram,

$$\tan \theta = \frac{qr}{oq} = \frac{qr}{po + qp} = \frac{I_0 \cdot \sin[90^\circ - (\delta + \alpha)]}{nI_S + I_0 \cdot \cos[90^\circ - (\delta + \alpha)]}$$

$$= \frac{I_0 \cdot \cos(\delta + \alpha)}{nI_S + I_0 \cdot \sin(\delta + \alpha)}$$

For very small angles, $\theta \approx \frac{I_0 \cdot \cos(\delta + \alpha)}{nI_S + I_0 \cdot \sin(\delta + \alpha)}$

This expression can still be simplified with the assumption $I_0 \ll nI_S$.

$$\theta \approx \frac{I_0 \cos(\delta + \alpha)}{nI_S} \text{ rad}$$

Or, $\theta \approx \frac{I_0 (\cos \delta \cos \alpha - \sin \delta \sin \alpha)}{nI_S} \approx \frac{I_M \cos \delta - I_C \sin \delta}{nI_S} \text{ rad}$

$$\text{Or, phase angle of CT } \theta \approx \frac{180}{\pi} \left(\frac{I_M \cos \delta - I_C \sin \delta}{nI_S} \right) \text{ degree}$$

ERRORS INTRODUCED BY CT:-

1. RATIO ERROR:

$$\text{Percentage ratio error} = \frac{\text{Nominal ratio} - \text{Actual ratio}}{\text{Actual ratio}} \times 100\%$$

$$= \frac{Kn - R}{R} \times 100\%$$

where, Nominal ratio = $\frac{\text{Rated primary winding current}}{\text{Rated secondary winding current}} = Kn \approx n$

In practice, the CT burden is largely resistive with a small value of inductance, thus the secondary phase angle δ is positive and generally small. The nominal ratio Kn , is sometimes loosely taken equal to the turns ratio n .

Thus, $\sin \delta \approx 0$ and $\cos \delta \approx 1$. Therefore, Eq. (3.3) can be approximated as

$$R \approx n + \frac{I_M \sin \delta + I_C \cos \delta}{I_S} \approx n + \frac{I_C}{I_S}$$

Accordingly, percentage ratio error can be approximated as

$$\text{Percentage ratio error} = \frac{Kn - \left(n + \frac{I_C}{I_S} \right)}{\left(n + \frac{I_C}{I_S} \right)} \times 100\%$$

2. PHASE ANGLE ERROR:

Error in phase angle is given following Eq. (3.6) as

$$\theta \approx \frac{180}{\pi} \left(\frac{I_M \cos \delta - I_C \sin \delta}{nI_S} \right)$$

In practice, the CT burden is largely resistive with a small value of inductance, thus the secondary phase angle δ is positive and generally small.

Thus, $\sin \delta \approx 0$ and $\cos \delta \approx 1$. Therefore, Eq. (3.10) can be approximated as

$$\theta \approx \frac{180}{\pi} \times \frac{I_M}{nI_S} \text{ degree}$$

CAUSES OF ERRORS:

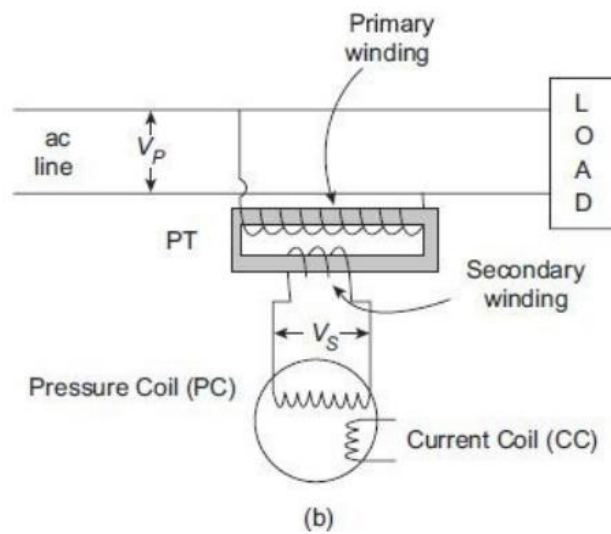
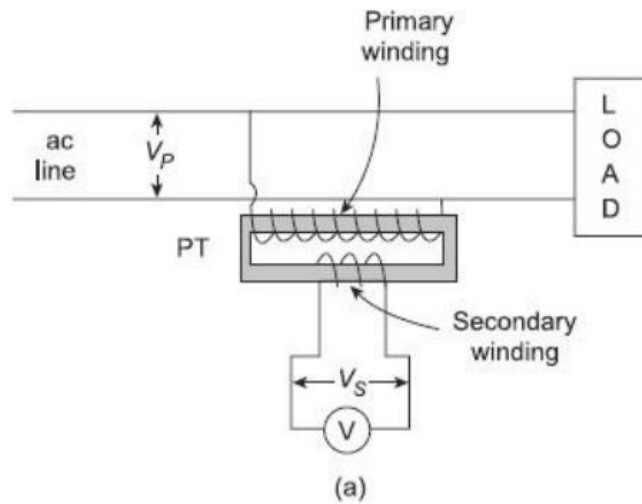
In an ideal CT, the actual transformation ratio should have been exactly equal to the turns ratio and the phase angle should have been zero. However, due to inherent physical limitations inherent to the electric and magnetic circuits of the CT, practical performance deviates from these ideal behaviours and errors are introduced in measurement.

1. Primary winding always needs some magnetising MMF to produce flux and, Therefore, the CT draws the magnetising current I_M .
2. CT no-load current must have a component I_C that has to supply the Core losses, i.e., The eddy current loss and the hysteresis loss.
3. Once the CT core becomes saturated, the flux density in the core no longer remains a Linear function of the magnetising force, this may introduce further errors.
4. Primary and secondary flux linkages differ due to unavoidable flux leakages.

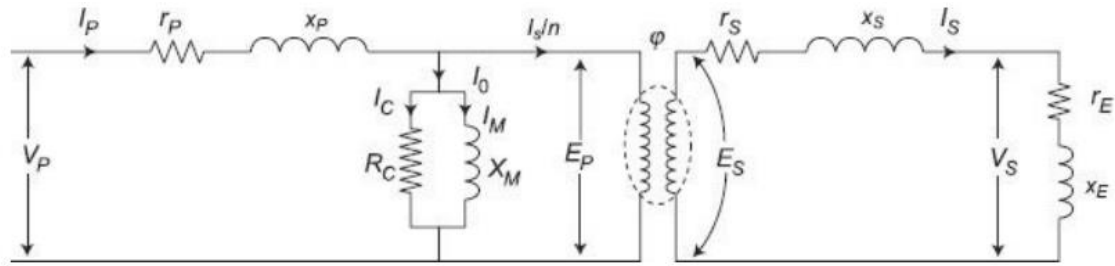
REDUCING ERRORS IN CT:

1. Low flux density
2. High permeability core material
3. Modification of turns ratio
4. Use of shunts
5. Wound-core construction

POTENTIAL TRANSFORMER (PT)



Use of PT for (a) voltage, and (b) power measurement



3.15 Equivalent circuit of a PT

V_P = primary supply voltage

E_P = primary winding induced voltage

V_S = secondary terminal voltage

E_S = secondary winding induced voltage

I_P = primary current

I_S = secondary current

I_0 = no-load current

I_C = core loss component of current

I_M = magnetising component of current

r_P = resistance of primary winding

x_P = reactance of primary winding

r_S = resistance of secondary winding

x_S = reactance of secondary winding

R_C = imaginary resistance representing core losses

X_M = magnetising reactance

r_E = resistance of external load (burden) including resistance of meters, current coils etc.

x_E = reactance of external load (burden) including reactance of meters, current coils, etc.

N_P = primary winding number of turns

N_S = secondary winding number of turns

n = turns ratio

$$= \frac{N_P}{N_S}$$

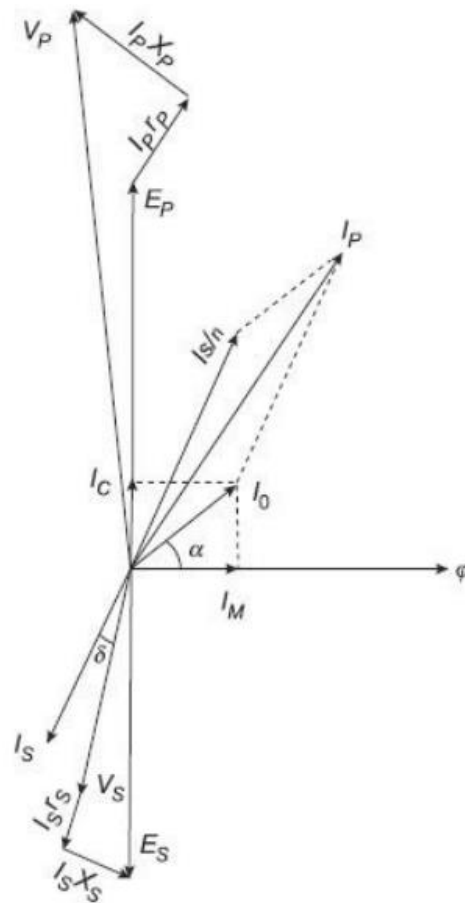
ϕ = working flux of the PT

θ = the 'phase angle' of the PT

δ = phase angle between secondary winding terminal voltage and secondary winding current (i.e., phase angle of load circuit)

β = phase angle between primary load current and secondary terminal voltage reversed

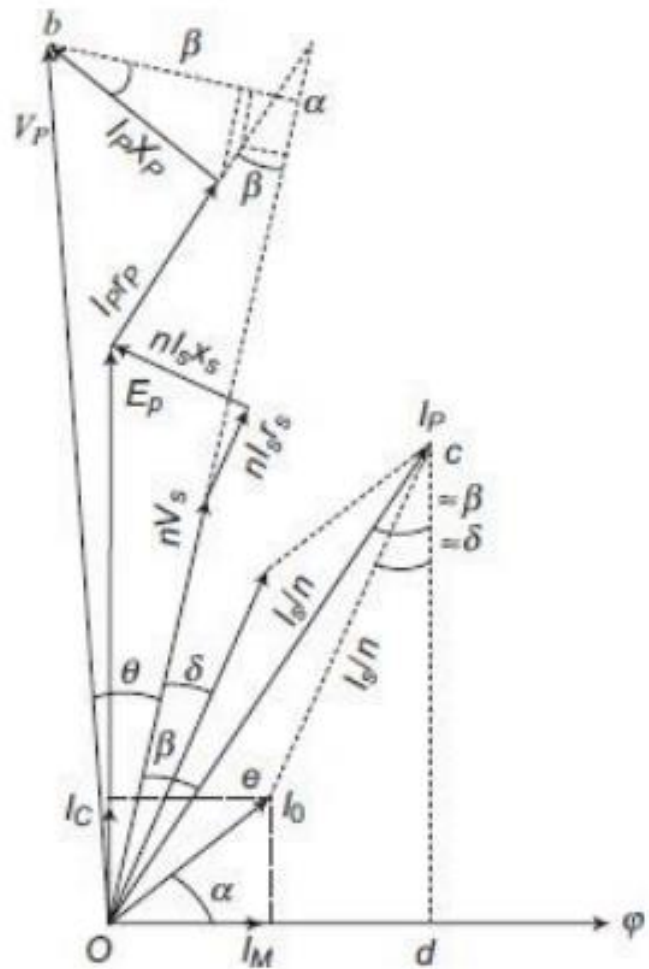
α = phase angle between no-load current I_0 and flux ϕ



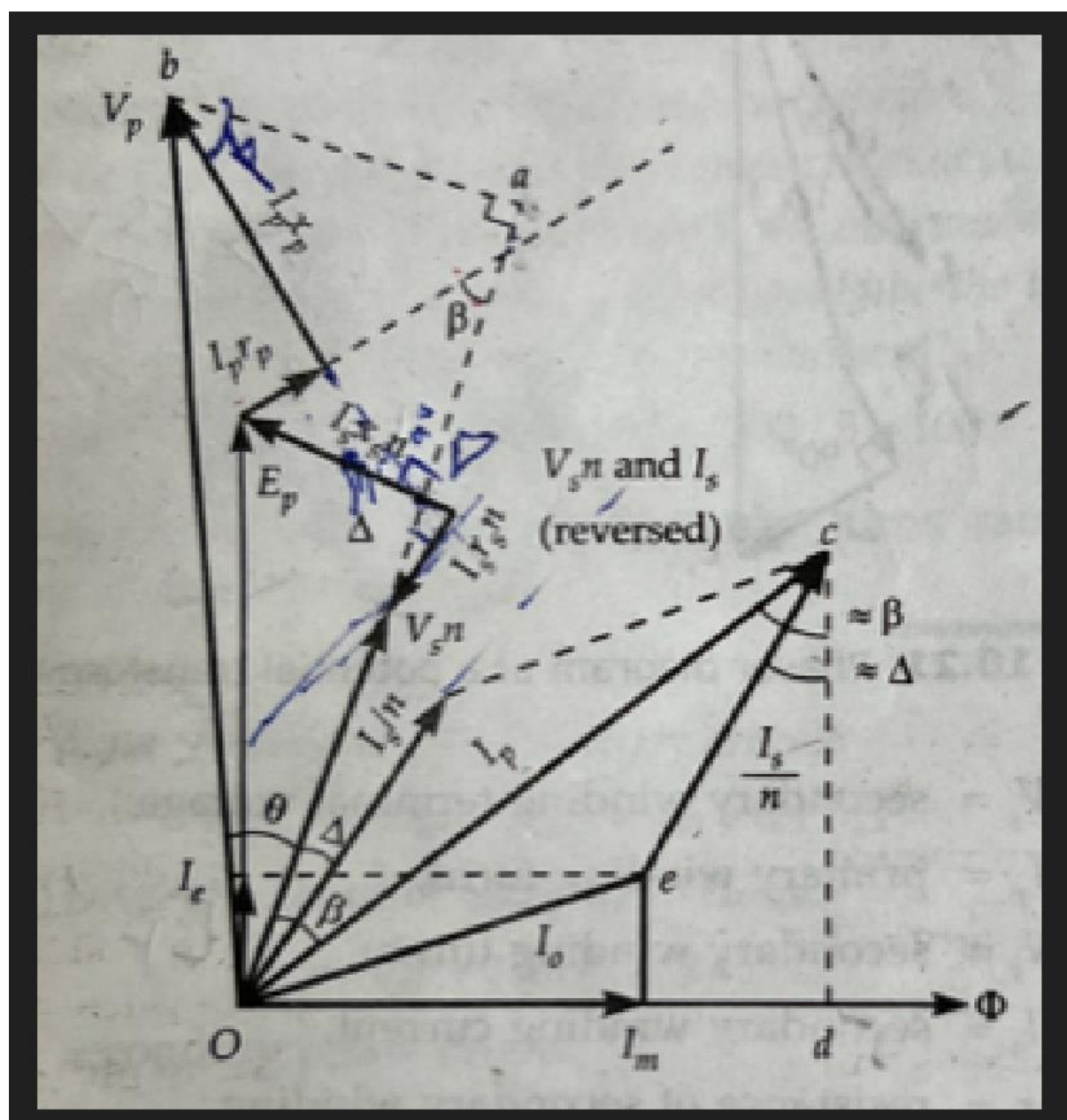
6 Phasor diagram of a PT

Secondary voltages when referred to primary side need to be multiplied by the turns ratio n , whereas, when secondary currents are to be referred to primary side, they need to be divided by n . Secondary current I_S , when reflected back to primary, can be represented by the 180° shifted phasor indicated by I_S/n . Primary winding current I_P is the phasor summation of this reflected secondary current (load component) I_S/n and the no-load current I_0 .

VOLTAGE TRANSFORMATION RATIO OF PT:



7 Expanded view of a section of Figure 3.16



$$\overline{V_p} = \overline{E_p} + \overline{I_p r_p} + \overline{I_p x_p} = n\overline{E_s} + \overline{I_p r_p} + \overline{I_p x_p}$$

$$\text{or, } \overline{V_p} = n(\overline{V_s} + \overline{I_s r_s} + \overline{I_s x_s}) + \overline{I_p r_p} + \overline{I_p x_p} = n\overline{V_s} + n\overline{I_s r_s} + n\overline{I_s x_s} + \overline{I_p r_p} + \overline{I_p x_p}$$

From the phasor diagram in Figure 3.17, we have:

$$oa = V_p \cos \theta$$

and

$$oa = nV_s + nI_s r_s \cos \delta + nI_s x_s \sin \delta + I_p r_p \cos \beta + I_p x_p \sin \theta$$

$$\text{or, } V_p \cos \theta = nV_s + nI_s r_s \cos \delta + nI_s x_s \sin \delta + I_p r_p \cos \beta + I_p x_p \sin \beta$$

$$\text{or, } V_p \cos \theta = nV_s + nI_s (r_s \cos \delta + x_s \sin \delta) + I_p r_p \cos \beta + I_p x_p \sin \beta \quad (3.12)$$

In reality, the phase angle difference θ is quite small, thus for the sake of simplicity, both V_p and V_s reversed can be approximated to perpendicular to the flux $\mathbf{j}$, and hence:

$$\angle ocd \approx \beta \text{ and } \angle ecd \approx \delta$$

$$\text{Thus, } I_p \cos \beta = I_C + \frac{I_s}{n} \cos \delta \text{ and } I_p \sin \beta = I_M + \frac{I_s}{n} \sin \delta \quad (3.13)$$

In reality, once again, since θ is very small, sometimes even less than 1° , then we can approximate as

$$\cos \theta = 1, \text{ and } V_p \cos \theta = V_p$$

Substituting the above values in Eq. (3.12) we have,

$$V_p = nV_s + nI_s (r_s \cos \delta + x_s \sin \delta) + \left(I_C + \frac{I_s}{n} \cos \delta \right) r_p + \left(I_M + \frac{I_s}{n} \sin \delta \right) x_p$$

$$\text{or, } V_p = nV_s + I_s \cos \delta \left(nr_s + \frac{r_p}{n} \right) + I_s \sin \delta \left(nx_s + \frac{x_p}{n} \right) + (I_C r_p + I_M x_p) \quad (3.14)$$

$$\text{or, } V_p = nV_s + \frac{I_s}{n} \cos \delta (n^2 r_s + r_p) + \frac{I_s}{n} \sin \delta (n^2 x_s + x_p) + (I_C r_p + I_M x_p)$$

$$\text{or, } V_p = nV_s + \frac{I_s}{n} \cos \delta R_p + \frac{I_s}{n} \sin \delta X_p + (I_C r_p + I_M x_p)$$

$$\text{or, } V_p = nV_s + \frac{I_s}{n} (R_p \cos \delta + X_p \sin \delta) + (I_C r_p + I_M x_p) \quad (3.15)$$

Here, R_p = equivalent resistance of the PT referred to primary side

X_p = equivalent reactance of the PT referred to primary side

Thus, actual voltage transformation ratio:

$$R = \frac{V_p}{V_s} = n + \frac{\frac{I_s}{n} (R_p \cos \delta + X_p \sin \delta) + (I_C r_p + I_M x_p)}{V_s} \quad (3.16)$$

Equation (3.14) may be re-written as

$$V_p = nV_s + nI_s \cos \delta \left(r_s + \frac{r_p}{n^2} \right) + nI_s \sin \delta \left(x_s + \frac{x_p}{n^2} \right) + (I_C r_p + I_M x_p)$$

or, $V_p = nV_s + nI_s \cos \delta R_s + nI_s \sin \delta X_s + (I_C r_p + I_M x_p)$

or, $V_p = nV_s + nI_s (R_s \cos \delta + X_s \sin \delta) + (I_C r_p + I_M x_p)$ (3.17)

Here, R_s = equivalent resistance of the PT referred to secondary side

X_s = equivalent reactance of the PT referred to secondary side

Thus, actual voltage transformation ratio can again be written as

$$R = \frac{V_p}{V_s} = n + \frac{nI_s (R_s \cos \delta + X_s \sin \delta) + (I_C r_p + I_M x_p)}{V_s} \quad (3.18)$$

Following Eqs (3.16) and (3.18), the error in ratio, i.e., the difference between actual transformation ratio and turns ratio can be expressed in either of the following two

forms: $R - n = \frac{\frac{I_s}{n} (R_p \cos \delta + X_p \sin \delta) + (I_C r_p + I_M x_p)}{V_s} \quad (3.19)$

$$= \frac{nI_s (R_s \cos \delta + X_s \sin \delta) + (I_C r_p + I_M x_p)}{V_s} \quad (3.20)$$

PHASE ANGLE OF PT:

From the phasor diagram of [Figure 3.17](#),

$$\tan \theta = \frac{ab}{oa} = \frac{I_P x_P \cos \beta - I_P r_P \sin \beta + n I_S}{n V_S + n I_S r_S \cos \delta + n I_S x_S \sin \delta}$$

To simplify the computations, here we can make the assumption that in the denominator, the terms containing I_P and I_S being much less compared to the large voltage nV_S , those terms can be neglected; thus we get a simplified form:

$$\tan \theta = \frac{I_P x_P \cos \beta - I_P r_P \sin \beta + n I_S x_S \cos \delta - n I_S r_S \sin \delta}{n V_S} \quad (3.21)$$

Following Eq. (3.12), we get

$$\begin{aligned} \tan \theta &= \frac{x_P \left(I_C + \frac{I_S}{n} \cos \delta \right) - r_P \left(I_M + \frac{I_S}{n} \sin \delta \right) + n I_S x_S \cos \delta - n I_S r_S \sin \delta}{n V_S} \\ \text{or, } \tan \theta &= \frac{I_S \cos \delta \left(\frac{x_P}{n} + n x_S \right) - I_S \sin \delta \left(\frac{r_P}{n} + n r_S \right) + I_C x_P - I_M r_P}{n V_S} \\ \text{or, } \tan \theta &= \frac{\frac{I_S \cos \delta}{n} (x_P + n^2 x_S) - \frac{I_S \sin \delta}{n} (r_P + n^2 r_S) + I_C x_P - I_M r_P}{n V_S} \\ \text{or, } \tan \theta &= \frac{\frac{I_S \cos \delta}{n} X_P - \frac{I_S \sin \delta}{n} R_P + I_C x_P - I_M r_P}{n V_S} \\ \text{or, } \tan \theta &= \frac{\frac{I_S}{n} (X_P \cos \delta - R_P \sin \delta) + I_C x_P - I_M r_P}{n V_S} \end{aligned}$$

Since θ is small, we can assume $\tan \theta = \theta$; thus,

$$\theta = \frac{\frac{I_S}{n} (X_P \cos \delta - R_P \sin \delta) + I_C x_P - I_M r_P}{n V_S} \quad (3.22)$$

$$\text{or, } \theta = \frac{\frac{I_S}{n} (n^2 X_S \cos \delta - n^2 R_S \sin \delta) + I_C x_P - I_M r_P}{n V_S}$$

$$\text{or, } \theta = \frac{n I_S (X_S \cos \delta - R_S \sin \delta) + I_C x_P - I_M r_P}{n V_S}$$

$$\text{Thus, phase angle } \theta = \frac{I_S}{V_S} (X_S \cos \delta - R_S \sin \delta) + \frac{I_C x_P - I_M r_P}{n V_S} \quad (3.23)$$

ERRORS INTRODUCED BY POTENTIAL TRANSFORMERS:

1. Ratio Error
2. Phase-Angle Error

REDUCING ERRORS IN PT:

1. Reducing the loss component and magnetising components, i.e., the no-load component of the primary current can be achieved by reducing the length of magnetic path in the core, using good quality core magnetic materials, designing with appropriate value of flux densities in the core, and adopting precautionary measures while assembling and interleaving of core laminations.
2. Winding resistance can be reduced by using thick conductors and taking care to reduce the length of mean turn of the windings.
3. Winding leakage flux and hence leakage reactance can be reduced by keeping the primary and secondary windings as close as permissible from the point of view of insulation requirements.
4. Sufficiently high flux densities in the core will reduce the core cross-section, thereby reducing the length of winding wound over the core. This, in turn, will reduce the winding resistance. An optimisation in the core flux density value to be used needs to be done, since too high a flux density will increase the no-load current, which is also not desirable.